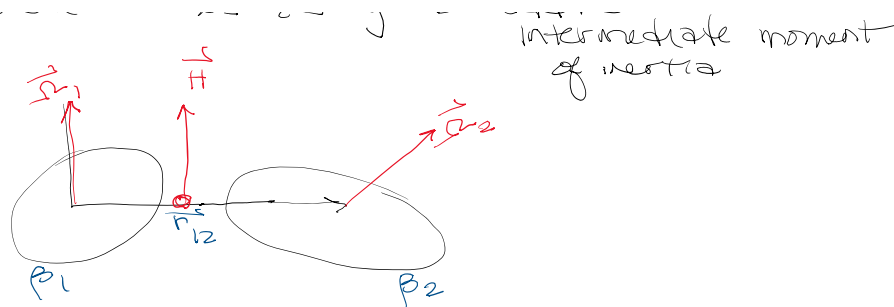


$N=2$



intermediate moment of inertia

$$U = U_{12} + U_{11} + U_{22} \quad I_H = \frac{m_1 m_2}{m_1 + m_2} r_{12}^2 + \hat{H} \cdot \bar{A}_1 \bar{I}_1 \bar{A}_1^T \cdot \hat{H} + \hat{H} \cdot \bar{A}_2 \bar{I}_2 \bar{A}_2^T \cdot \hat{H}$$

"sphere restricted" 2BP

self potentials of sphere

$$U_{12} = -\frac{G m_1 m_2}{r_{12}} \quad U_{11} = -\frac{3}{5} \frac{G M_1^2}{R_1} \quad U_{22} = -\frac{3}{5} \frac{G M_2^2}{R_2} \quad r_{12} \geq R_1 + R_2$$

$$I_H = \frac{m_1 m_2}{m_1 + m_2} r_{12}^2 + \underbrace{\frac{2}{5} M_1 R_1^2 + \frac{2}{5} M_2 R_2^2}_{I_S} \quad \text{must keep the } I_S \text{ around}$$

$H \Rightarrow$ includes spins of β_1 & β_2

Where are torques? Grav torques = 0



total effects

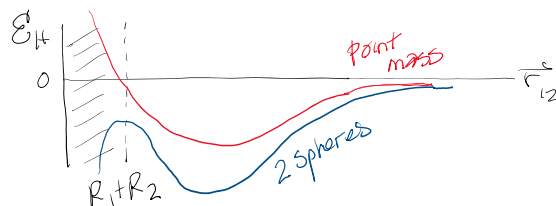


spin rates will change over time

$$\mathcal{E}_H = \frac{H^2}{2 \frac{m_1 m_2}{m_1 + m_2} r_{12}^2 + I_S} - \frac{G m_1 m_2}{r_{12}} + (U_{11} + U_{12})$$

$$\mathcal{E}_H(r_{12}, \theta)$$

plot this as func of radius



Relative Equil occurs @

$$\delta \mathcal{E}_H \geq 0$$

$$\delta \mathcal{E}_H = \frac{\partial \mathcal{E}_H}{\partial r_{12}} \delta r_{12}$$

for all admissible deviations δr_{12}

$$\frac{\partial \mathcal{E}_H}{\partial r_{12}} = 0 \Rightarrow \text{orbital case of equilibrium}$$



What if $r_{12} = R_1 + R_2$



$\delta r_{12} \geq 0$ only

$\delta r_{12} < 0$ impossible

$$\frac{\partial \mathcal{E}_H}{\partial r_{12}} \propto -F_{12}$$

$$\left(\frac{\partial \mathcal{E}_H}{\partial r_{12}} \right) (\delta r_{12}) \Rightarrow \text{if } \frac{\partial \mathcal{E}_H}{\partial r_{12}} \cdot \delta r_{12} \geq 0 \text{ for all } \delta r_{12} \geq 0$$

relative equilibrium

$$\underbrace{0, 1, 2, \dots}_{-F_{12} > 0 \geq 0}$$

relative equilibrium

If H^2 is large enough then $\frac{\partial \mathcal{E}_H}{\partial r_{12}} < 0$

$$-\Omega^2 r_{12} - F_{12} < 0 \quad \text{if } \Omega^2 \text{ large enough}$$

fission of system

Given a config space $\mathcal{Q} = \{r_{ij}, \bar{A}_i\}$ (generality)

$$\mathcal{E}_H(\mathcal{Q}) = \frac{H^2}{2I_H(\mathcal{Q})} + U(\mathcal{Q})$$

relative equilibria exist if $\delta \mathcal{E}_H \geq 0 \quad \forall$ possible $\delta \mathcal{Q} \geq 0$

$$\delta \mathcal{E}_H \geq 0 = \begin{cases} \frac{\partial \mathcal{E}_H}{\partial q_i} = 0 & \text{for all } q_i \text{ "orbital case"} \\ \text{some } \frac{\partial \mathcal{E}_H}{\partial q_i} = 0 \\ \frac{\partial \mathcal{E}_H}{\partial q_j} \delta q_j \geq 0 & \text{for } \delta q_j \geq 0 \text{ "contact equilibria"} \end{cases}$$

general condition for equl.

Stability (Energetic)

For orbital conditions, $\frac{\partial \mathcal{E}_H}{\partial q_i} = 0$, holds for $i = 1, 2, \dots, M$

Then $\delta \mathcal{E}_H > 0$ for $\delta q_i = 1, 2, \dots, M$ arbitrary
 $\delta \mathcal{E}_H \geq 0$ at the constrained $\delta q_j \geq 0$
 $j = M+1, \dots, N$

determines stability

Non dimensionalize the 2BP

$$r = \frac{r_{12}}{R_1 + R_2} \geq 1$$

$$\bar{I}_0 = \frac{m_1 m_2}{m_1 m_2 (R_1 + R_2)^2} I_0 = \frac{2}{5} \left[\frac{1}{\mu_1} r_1^2 + \frac{1}{\mu_2} r_2^2 \right]$$

$$\bar{r}_1 = \frac{R_1}{R_1 + R_2}$$

$$\mu = \frac{m_1}{m_1 + m_2}$$

$$d_{12} = R_1 + R_2$$

$$\bar{H} = \frac{\sqrt{m_1 + m_2}}{\sqrt{G d_{12} m_1 m_2}} H$$

$$\bar{U} = \frac{d_{12}}{G m_1 m_2} U$$

$$\bar{\mathcal{E}}_H = \frac{\bar{H}^2}{2(r^2 + \bar{I}_0)} - \frac{1}{r}$$

ignore U_{11}, U_{22}

$$\mathcal{E}_H = \frac{H^2}{2(r^2 + I_0)} - \frac{1}{r}$$

DOF is r

Relative Equil $\delta \mathcal{E}_H = \frac{\partial \mathcal{E}_H}{\partial r} \delta r = \left[\frac{-H^2}{2(r^2 + I_0)} (2r) + \frac{1}{r^2} \right] \delta r \geq 0$

$$(r^2 + I_0)^2 - r^3 H^2 = 0$$

$$r^4 - H^2 r^3 + 2 I_0 r^2 + I_0 = 0$$

Descartes rule of signs 2 or 0 real roots

$$\text{if } H^2 < 1$$

For H^2 small enough we only have resting equl...

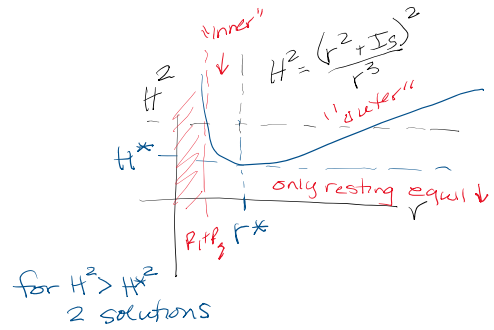
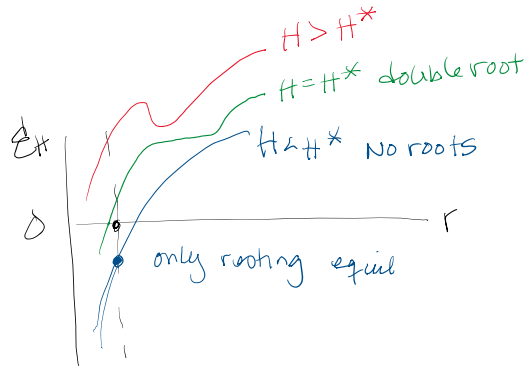
$$\exists \text{ a double root at some value of } H^2 \quad H^2 = \frac{(r^2 + I_0)^2}{r^3}$$

Double root condition will be $[4r^2 - 3H^2 r + 4I_0] r = 0$

$$\Rightarrow r^* = \sqrt{3I_0} \quad H^{*2} = \frac{16}{3\sqrt{3}} \sqrt{I_0}$$

$$\Rightarrow r^* = \sqrt{3I_s} \quad H^{*2} = \frac{16}{3\sqrt{3}} \sqrt{I_s}$$

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Consider stability of the solutions (2nd variation)

$$\delta^2 E_H = \left(\frac{\partial^2 E_H}{\partial r^2} \right) (\delta r)^2 = \left[\frac{r^2 - 3I_s}{r^3(r^2 + I_s)} \right] (\delta r)^2$$

if $r^2 < 3I_s \Rightarrow \delta^2 E_H < 0$ energetically unstable

if $r^2 > 3I_s \Rightarrow \delta^2 E_H > 0$ stable

Contact stability

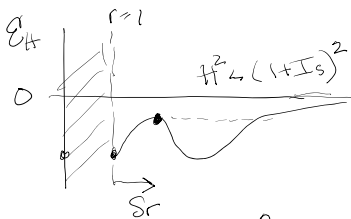
$$\delta E_H = \left[\frac{-H^2 r}{(r^2 + I_s)^2} - \frac{1}{r^2} \right] \delta r \geq 0 \quad \forall \delta r \geq 0$$

$r = d_{12} = 1$

$$\left. \frac{\partial E_H}{\partial r} \right|_{r=1} = \left[\frac{-H^2}{(1+I_s)^2} + 1 \right] \geq 0 \quad \text{being held together}$$

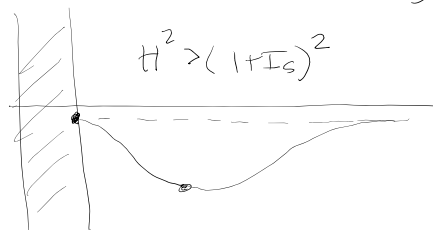
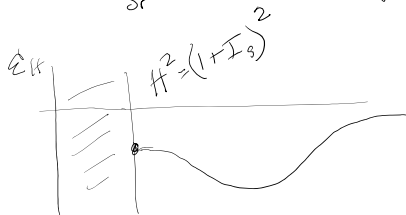
$\Rightarrow$ resting is stable as long as $H^2 \leq (1+I_s)^2$ when $H^2 = (1+I_s)^2$ contact force $\rightarrow 0$

Lecture 13 2/25/26



$$\left. \frac{\partial E_H}{\partial r} \right|_{r=1} = 0$$

when $H^2 = (1+I_s)^2 \Rightarrow$ contact force is zero



if H large enough system can't stay in contact

Can show that at $H^2 = (1+I_s)^2$ the rotation of the system (dimensional) is

$$\Omega^2 = \frac{G(m_1 + m_2)}{d_{12}^3}$$

by

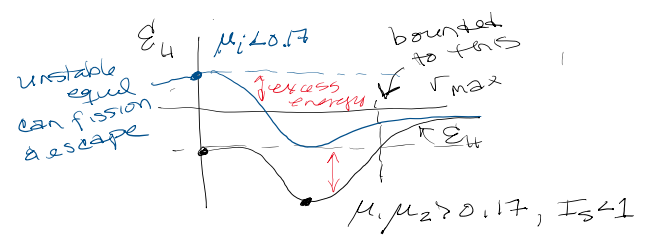
$$d_{12}$$

Let's look @ system energy when fission occurs

$$E_H \Big|_{r=1} = \frac{H^2}{2I_H} + U \Big|_{r=1} = \boxed{\frac{1}{2}(I_S - 1)} = \text{energy @ fission rate}$$

$$H^2 = (1 + I_S)^2$$

If $I_S > 1 \Rightarrow$ fission energy is positive, $I_S = \frac{2}{5} \left[\frac{(R_1/d_{12})^2}{\mu_2} + \frac{(R_2/d_{12})^2}{\mu_1} \right]$
 $I_S > 1$ when $\mu_1 \sim \leq 0.17$ or $\mu_2 \sim \leq 0.17$



Asteroid pairs - can integrate back in time & determine sometime in the past they were at same location w/ similar speeds. Most measured aside by $\mu < 0.17$. Excess energy generally comes from primary